

TEKNAF

ENVIRONMENTAL IMPACT ASSESSMENT

Teknaf to Tetulia Integrated Economic Corridor

North-West Dhaka–South-East Economic Corridor / Bangladesh

~700 km

44

US\$79 B

Coastal-to-Himalayan foothill corridor

Districts within the study belt

Proposed 20-year investment envelope

TETULIA

What this assessment covers

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02	Purpose of the Project Development rationale and strategic objectives	08	Socio-Economic & Environmental Impact Combined effects on people and ecosystems
03	Study Area & Corridor Map Regional footprint and ecological sensitivity	09	Nature-Based Solutions Ecosystem-based mitigation recommendations
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PROJECT DESCRIPTION

An integrated corridor from the Bay of Bengal to the Himalayan foothills

The Teknaf–Tetulia Integrated Economic Corridor is a proposed 20-year, ADB-supported development programme linking Bangladesh's southeastern coast to its northernmost tip. The alignment follows the existing Teknaf–Chattogram–Dhaka–Bogura–Rangpur–Tetulia transport spine, with development activities spanning a belt roughly 50 km on either side of the route, across 44 districts.

RAIL

Multi-modal spine

Highway, rail and river-port linkages connecting Matarbari, Chattogram, Dhaka, Gazipur, Bogura, Rangpur and Dinajpur.

SEZ

Industrial nodes

Special economic zones, logistics hubs and agro-processing clusters staged along the corridor.

LAND

Diverse terrain

Coastal wetlands, hill forests, floodplains, urban delta and Barind tract within a single alignment.

TIME

Phased delivery

US\$20 B (years 1–3), US\$45 B (years 4–10), remainder over the final decade.

PROJECT SNAPSHOT

PROPONENT

Government of Bangladesh, Economic Relations Division

DEVELOPMENT PARTNER

Asian Development Bank (ADB)

SOUTHERN TERMINUS

Teknaf, Cox's Bazar District

NORTHERN TERMINUS

Tetulia, Panchagarh District

INDICATIVE LENGTH

≈ 700 km alignment

STUDY BELT WIDTH

≈ 100 km (50 km either side)

DISTRICTS COVERED

44

PROGRAMME HORIZON

20 years

Why the corridor is being pursued

The corridor is intended to decongest Dhaka and Chattogram by distributing industrial growth, logistics and population pressure along a new north–south spine, while lifting lagging northern districts through improved connectivity. The Environmental Impact Assessment establishes the ecological and social baseline needed to steer that growth away from sensitive rivers, forests and coastal habitats.



Decongest primary cities

Redistribute industrial and demographic pressure away from Dhaka and Chattogram.



Raise national GDP growth

Targeted uplift of roughly 2 percentage points in average annual GDP growth.



Generate employment

An estimated 10 million jobs over the first decade of implementation.



Reduce poverty

Improved market access and services for an estimated 12 million people.



Strengthen northern agriculture

Rail and road links to move Rangpur-Dinajpur produce to national markets.

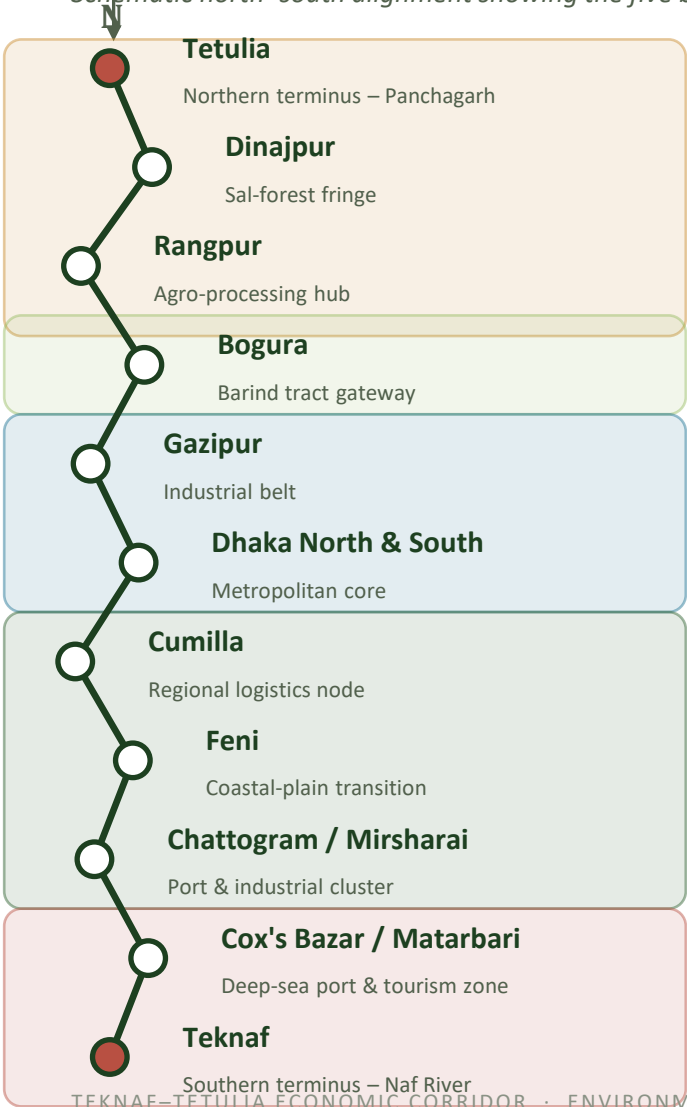


Improve public services

International-standard healthcare and logistics along the alignment.

Corridor study map — regional zones

Schematic north–south alignment showing the five broad planning zones within the ~100 km study belt.



PLANNING ZONES

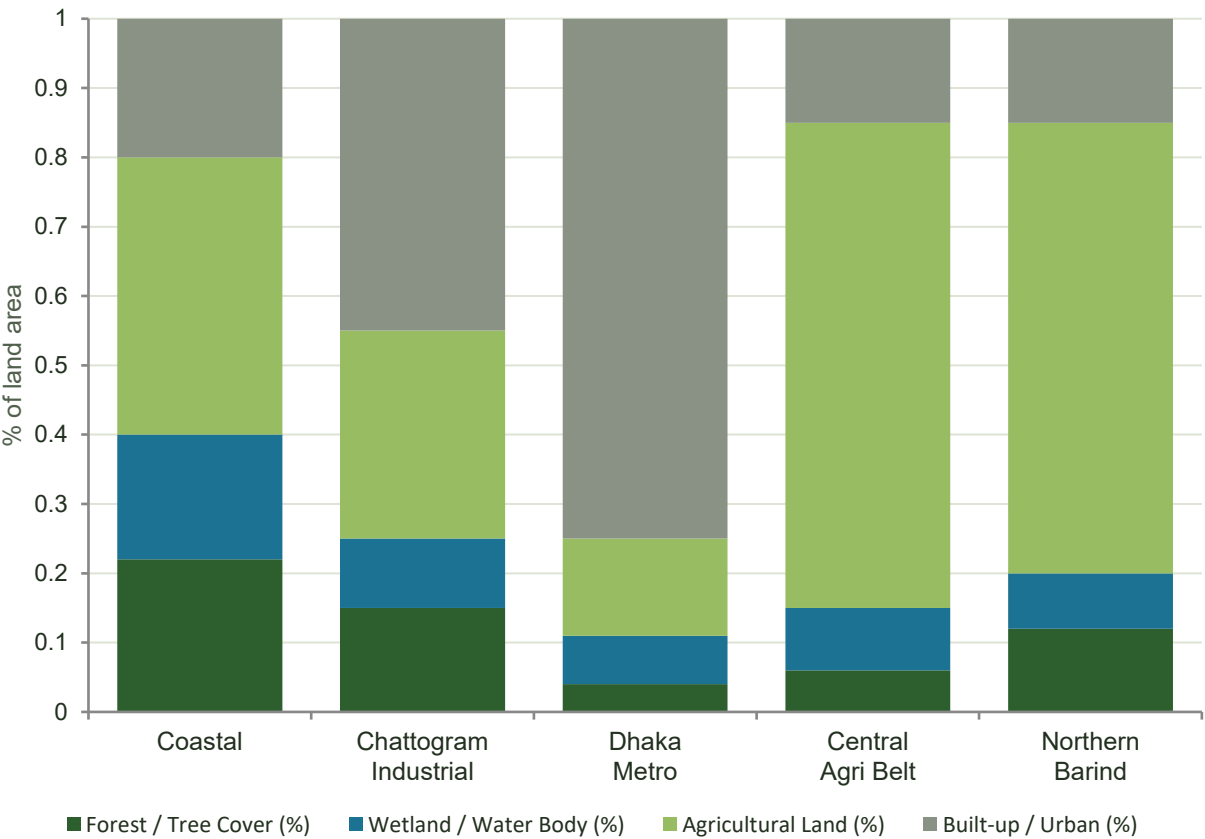
-  Northern / Barind Belt
-  Central Agricultural Belt
-  Dhaka Metropolitan Zone
-  Chattogram Industrial Belt
-  Coastal & Estuarine Zone

Development activities are examined within a 100 km belt (50 km either side of the alignment), spanning 44 districts.

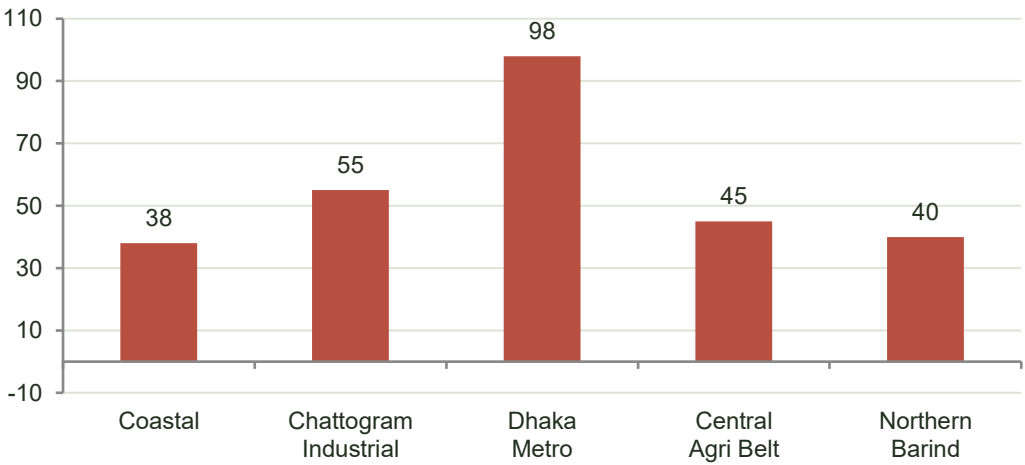
The corridor crosses three major river systems — the Naf, the Meghna estuary and the Jamuna — each requiring dedicated crossing and buffer design.

Existing conditions across the corridor zones

LAND-COVER COMPOSITION BY ZONE



AMBIENT AIR QUALITY — PM2.5 (µg/m³)



National standard: 35 µg/m³ · WHO guideline: 5 µg/m³ — Dhaka Metro and Chattogram Industrial zones already exceed national limits.

KEY BASELINE OBSERVATIONS

River water quality declines sharply through the Dhaka Metro reach (DO < 3 mg/L in dry season); coastal and Barind zones retain the highest surface-water and forest integrity.

Anticipated impacts by environmental receptor



Air Quality

Dust and vehicle/plant emissions during construction; long-term traffic and industrial emissions near urban nodes.



Water Resources

Sedimentation and effluent risk at river crossings; altered drainage and groundwater drawdown near urban segments.



Land & Soil

Conversion of agricultural and forest land; soil compaction, erosion and borrow-pit degradation.



Biodiversity

Habitat fragmentation across wildlife sanctuaries and the Naf estuary mangroves; disruption of the Halda River fish-spawning grounds.



Noise & Vibration

Construction and traffic noise affecting settlements, schools and wildlife corridors near sensitive reaches.



Waste & Materials

Construction debris, spoil disposal and borrow-material sourcing across a multi-terrain alignment.

Significance of impact by phase and receptor

Environmental Receptor	Key Sensitive Location	Construction	Operation
Air Quality	Dhaka Metro & Chattogram nodes	Medium	Medium
Surface Water & Hydrology	Naf, Meghna & Jamuna crossings	High	Medium
Biodiversity & Forests	Teknaf sanctuary, Chunati, Halda River	High	Medium
Land Use & Soil	Barind tract, coastal agricultural belt	Medium	Low
Noise & Vibration	Settlements near Gazipur–Dhaka reach	Medium	Low
Involuntary Resettlement	Urban and peri-urban alignment segments	High	Low
Cultural Heritage	Historic sites near Cumilla & Bogura	Low	Low
Waste & Spoil Management	Borrow areas and river-crossing works	Medium	Low

SIGNIFICANCE KEY

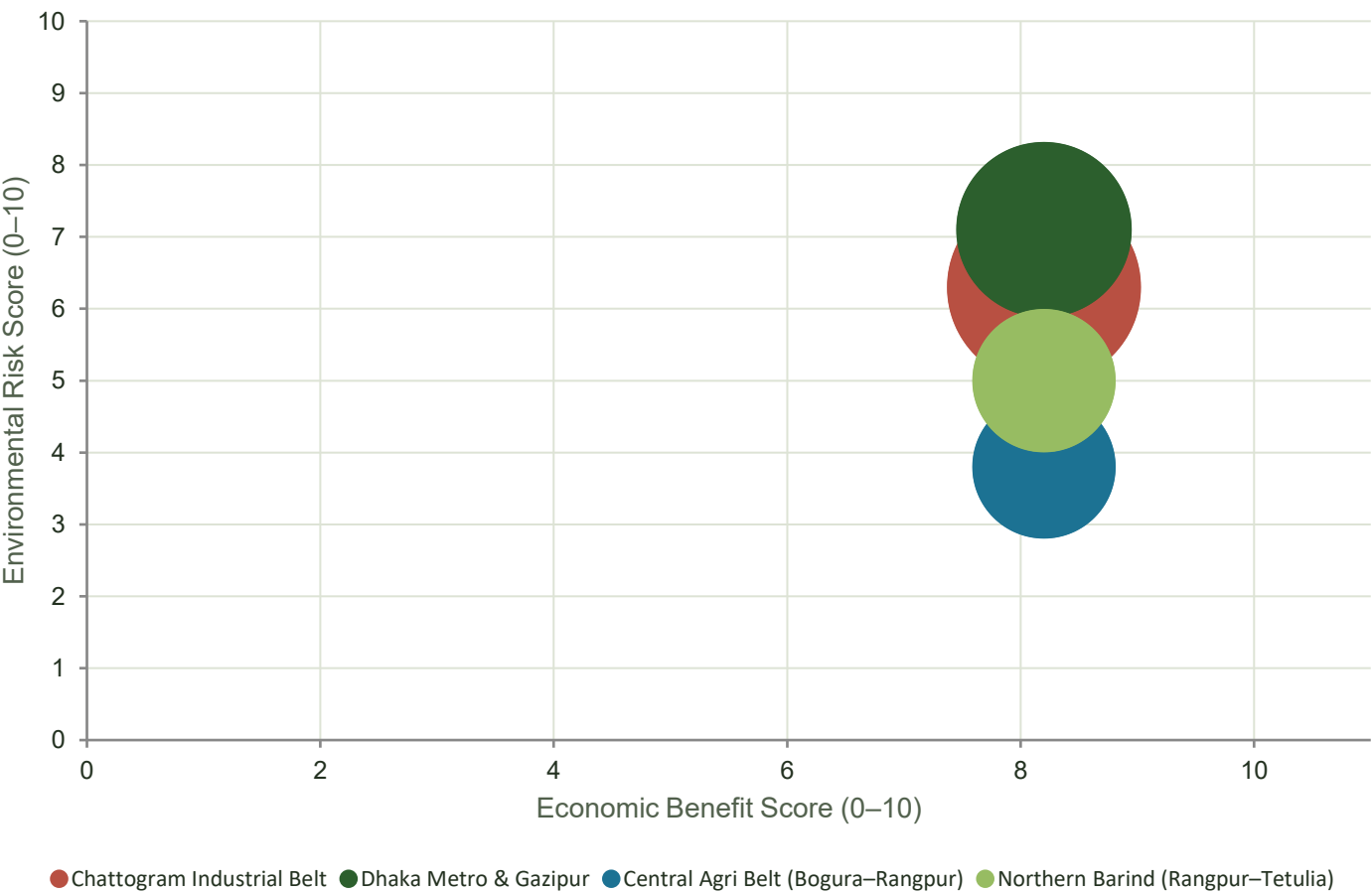
- Low — Minor, localised, reversible
- Medium — Moderate, manageable with mitigation
- High — Major, requires dedicated safeguards

READING THE MATRIX

Highest construction-phase significance concentrates on river crossings, biodiversity-rich reaches and resettlement along urban segments. Most impacts are rated moderate-to-low once mitigation and monitoring measures (Section 10) are applied during operation.

Projected environmental risk vs. economic benefit

By corridor segment — bubble size reflects indicative investment (US\$ billion)



HOW TO READ THIS

Segments in the upper-right quadrant deliver strong economic returns but carry material environmental exposure and should receive the most rigorous safeguard packages. Segments toward the lower-left are comparatively low-risk and can proceed with standard mitigation.

Coastal segment

Highest risk — mangrove and wildlife-sanctuary exposure at the Naf estuary; benefit driven by the Matarbari deep-sea port.

Chattogram Industrial

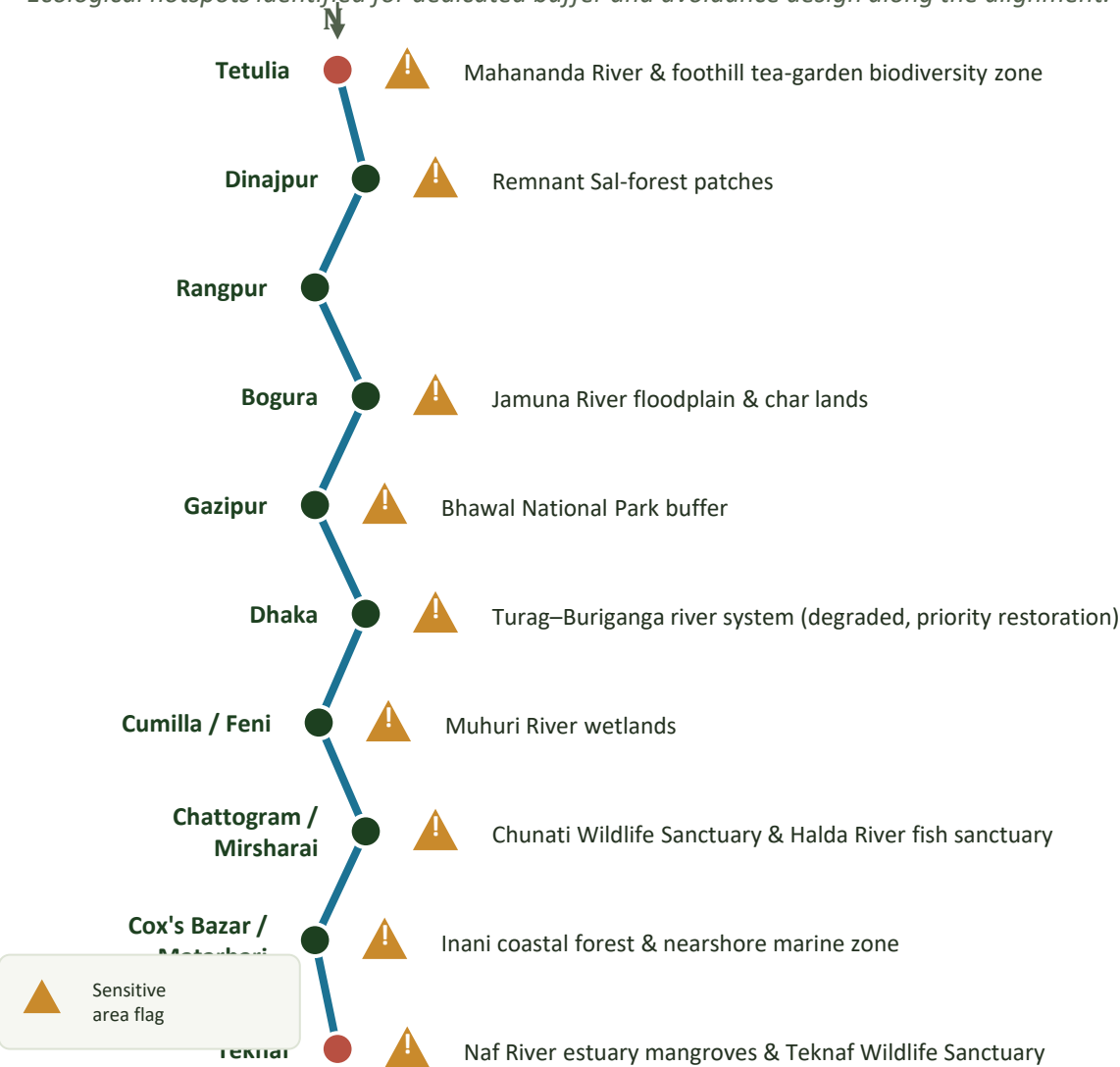
Best risk-adjusted return; Halda River spawning grounds need a dedicated buffer.

Dhaka Metro

Largest economic weight; risk stems from air quality and resettlement, not habitat loss.

Conceptual corridor route & environmentally sensitive areas

Ecological hotspots identified for dedicated buffer and avoidance design along the alignment.



Socio-economic and environmental impact

SOCIO-ECONOMIC IMPACT



Employment generation

~10 million jobs projected over the first implementation decade.



Market access for the north

Rail and road links open Rangpur–Dinajpur produce to national markets.



Poverty reduction

Improved connectivity and services for an estimated 12 million people.



Land acquisition & resettlement

Physical and economic displacement along urban and peri-urban segments.



Livelihood disruption

Temporary loss of access for farmers, fishers and small traders during construction.



In-migration pressure

Rapid population influx around new industrial nodes straining local services.

ENVIRONMENTAL IMPACT



Modal shift potential

Rail investment can reduce long-term per-tonne emissions versus road-only growth.



Planned green infrastructure

Corridor-wide tree belts and drainage designed in from the outset.



Habitat fragmentation

Wildlife sanctuaries and forest patches bisected by new alignment sections.



Hydrological disruption

River crossings and embankments alter flow, sediment and fish migration.



Land-use conversion

Agricultural and forest land converted to industrial and built-up use.



Cumulative air & water loading

Compounding emissions and effluent around Dhaka and Chattogram nodes.

Nature-based solutions along the corridor



NAF RIVER ESTUARY

Mangrove restoration buffer

Replant and protect a mangrove buffer to stabilise banks and preserve nursery habitat near Teknaf.



HALDA RIVER

Fish-passage & river buffer

Maintain natural flow and a construction-free buffer around Bangladesh's only natural carp spawning ground.



DHAKA METRO

Urban wetland retention

Retain and rehabilitate remaining wetlands and khals for stormwater attenuation and flood control.



FULL ALIGNMENT

Green corridor tree belt

Continuous roadside and rail-side planting for shade, dust capture, noise buffering and habitat linkage.



JAMUNA CROSSING

Floodplain-sensitive design

Elevated and permeable crossing design to preserve char-land flow paths and fish migration.



RANGPUR–DINAJPUR

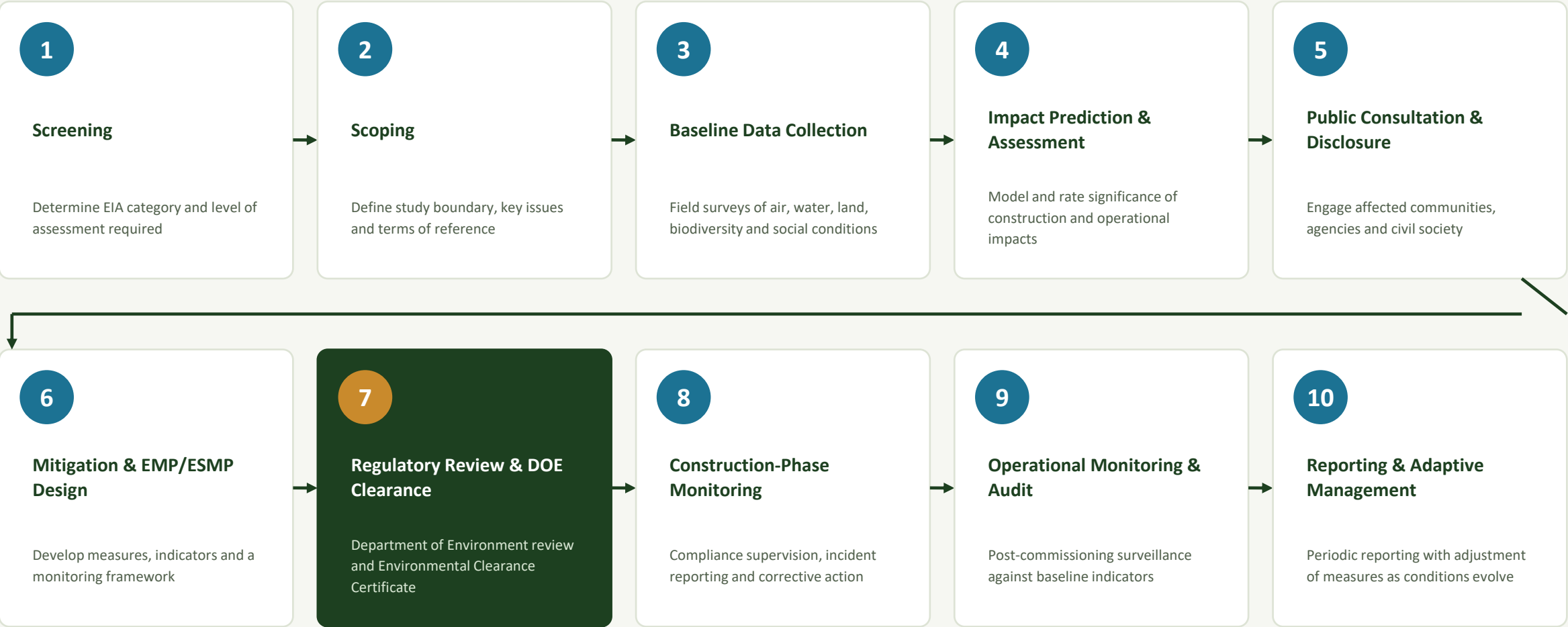
Barind agroforestry buffer

Community agroforestry belts to reduce erosion and support Sal-forest connectivity in the Barind tract.

Measures, indicators and responsibilities

Impact	Mitigation Measure	Monitoring Indicator	Responsible Agency	Frequency
Habitat loss (Naf estuary, Chunati)	Route avoidance/realignment; mangrove replanting; wildlife underpasses	Mangrove cover; sanctuary encroachment	Forest Dept. / DOE	Quarterly
River water quality	Silt curtains at crossings; treated construction effluent; no in-stream spoil disposal	Turbidity, DO, BOD at crossing points	DOE / WARPO	Monthly
Halda River fish spawning	Construction-free buffer and seasonal work restriction during spawning window	Spawning incidence; fish larvae counts	Dept. of Fisheries	Seasonal
Air quality (urban nodes)	Dust suppression, covered haulage, low-emission construction plant	PM2.5 / PM10 at sensitive receptors	DOE	Continuous
Resettlement & livelihoods	Resettlement Action Plan; livelihood restoration; grievance redress mechanism	Households compensated; grievances resolved	RHD / PIU Safeguards Unit	Monthly
Noise near settlements	Time-restricted works; noise barriers near schools and clinics	Ambient noise level (dB)	DOE / Contractor	Quarterly
Spoil & construction waste	Designated disposal sites; borrow-pit rehabilitation plan	Volume disposed vs. plan; site rehabilitation status	Contractor / PIU	Monthly

EIA process workflow



Institutional framework and regulatory alignment

INSTITUTIONAL ROLES

ERD	Economic Relations Division Overall programme ownership and ADB coordination
DOE	Department of Environment Environmental Clearance Certificate and compliance oversight
MoEFCC	Ministry of Environment, Forest & Climate Change Policy oversight and forest/wildlife clearances
RHD / BR	Roads & Highways Dept. / Bangladesh Railway Implementing agencies for physical works
PIU Safeguards Unit	Project Implementation Unit Day-to-day ESMP execution and grievance redress
Local Government	District & Upazila administrations Community liaison and local permitting

POLICY & SAFEGUARD ALIGNMENT

- Bangladesh Environment Conservation Act, 1995 (amended 2010) & Rules, 2023
- Environmental Conservation Rules — EIA categorisation and clearance procedure
- ADB Safeguard Policy Statement (2009): environment, resettlement, indigenous peoples
- Bangladesh Wildlife (Conservation & Security) Act, 2012
- National Water Policy and River-crossing design guidelines

GRIEVANCE REDRESS MECHANISM

A multi-tier GRM — community liaison, Upazila committee and PIU Safeguards Unit — will log, track and resolve environmental and social grievances raised by affected communities throughout construction and operation.

CONCLUSION

Growth and ecological integrity can advance together

With route avoidance at critical habitats, dedicated river buffers, nature-based solutions and a resourced monitoring programme, the Teknaf–Tetulia corridor can deliver its projected economic and social benefits while safeguarding the Naf estuary, Chunati and Halda ecosystems, the Barind tract and the corridor's river systems for the long term.

- 1 Adopt segment-specific safeguard packages guided by the risk-benefit assessment
- 2 Finalise ESMPs and secure DOE Environmental Clearance before mobilisation
- 3 Resource independent environmental and social monitoring for the full 20-year horizon